Sarah Mathew

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Computer Science student at Ontario Tech University focused on compilers and GPU systems, with hands-on experience building a statically-typed language with an LLVM backend and optimizing HIP compute kernels on AMD GPUs. Passionate about how code is transformed into fast machine instructions, and eager to grow as a compiler engineer.

Education

Ontario Tech University - Bachelor of Science (B.S.) in Computer Science (Software Development Specialization)
Sept. 2024 – May 2028

- **Coursework:** Data Structures & Algorithms, Compilers, Operating Systems, Systems Programmin, Computer Graphics
- **ACM Chapter Chair:** set strategic direction, organized women-in-tech workshops and networking events (2024–2026)

Seneca Polytechnic - Associate of Science (A.S.) - Computer Programming
Sept. 2023 – Dec. 2024

- **Coursework:** Object-Oriented Programming, Software Testing, Project Management, Web and Frameworks

Job Experience

Software Engineer — TiketFix May 2026 – Present
Client-intake platform for legal practices · ***React Native*** · ***TypeScript*** · ***Next.js*** · ***Node.js***

- Delivered frontend for 4 products, including a CRM, two client-facing websites, and an iOS/Android app, serving 300+ law firms across North America
- Enhanced user engagement by implementing core features like live chat, real-time lawyer-client messaging, and case submission and acceptance flows, contributing to a platform that has processed 80,000+ ticket submissions
- Streamlined client payments by integrating a Stripe-powered billing UI, enabling clients to pay lawyers directly through the platform

STEM Instructor — Exceed Robotics Oct. 2025 – Apr. 2026
C · ***Python*** · ***Arduino*** · ***OnShape***

- Strengthened technical skills for 40+ students aged 7–15 by teaching C, Python, and embedded systems through hardware-software projects on ESP32 and Arduino covering GPIO, PWM, and serial communication
- Increased hands-on engagement by leading weekly 3D printing workshops connecting software logic to physical hardware

Frontend & UX Developer — ReliablyME (Riipen) Jan. 2025 – Mar. 2025
Commitment-tracking platform · ***React.js*** · ***TypeScript*** · ***Figma***

- Delivered 4 core pages as sole developer in a 10-week engagement, owning the full pipeline from Figma wireframes to production React.js and TypeScript
- Improved user task efficiency by identifying and resolving UX friction points through moderated usability testing, leading to direct code-level UI enhancements

Project Experience

Lyra Compiler C++ · LLVM · x86-64

- Designed and built a **statically-typed compiled language in C++**, including a hand-written lexer, recursive descent parser, scoped symbol table, and static type checker with descriptive error messages
- Generated native x86-64 binaries by implementing an **LLVM IR backend** with first-class functions and **closure/upvalue capture**; optimized output with AST-level constant folding and dead code elimination plus LLVM mem2reg, GVN, and CFG simplification
- Ensured correctness and memory safety across **30+ test programs with zero Valgrind errors**, measuring and documenting IR instruction count reductions on benchmark programs

ROCm GPU Kernel Suite HIP · C++ · rocprof

- Implemented and profiled **GPU compute kernels in HIP**, including tiled matrix multiplication (LDS/shared memory), parallel reduction with warp shuffle, Blelloch prefix scan, and privatized histogram
- Improved performance over a naive global-memory baseline by optimizing **tiled matmul**, validating GFLOPS and L2 cache hit rate gains with rocprof
- Mapped kernel instructions to AMD GPU assembly by tracing the full **HIP → LLVM IR → AMDGPU ISA** compilation path in Compiler Explorer

Note Ease — AI Note-Taking App (MLH Best Use of AI in Education) MERN · TensorFlow · Google Cloud

- Won **MLH Best Use of AI in Education out of 200+ submissions** by shipping a full MERN stack app in 36 hours
- Enabled hands-free note-taking by building a real-time **speech-to-notes pipeline** with Google Cloud Speech-to-Text and Gemini, plus **MediaPipe hand-gesture recognition** for document formatting

Additional Information

Languages: C++, C, Python, Java, JavaScript, TypeScript, SQL, Shell, HTML/CSS
Compilers & Systems: LLVM IR, LLVM Pass Manager, ROCm/HIP
GPU & Graphics: HIP kernels, GPU memory hierarchy, OpenGL
Frameworks: Next.js, React, React Native, Angular, Node.js/Express, Flask, TensorFlow, MediaPipe
Tools: Git, Valgrind, CMake, Linux/Unix, Figma, PostgreSQL, MongoDB
Certifications: Meta Version Control · Meta Intro Mobile Dev · Meta JS
Hackathons: DeltaHacks XI, HawkHacks, ETHWomen, GDSC Guelph, Ctrl-Hack-Del, Ellehacks, Hack the 6ix